- **Rolling zone** is the area behind the paver that is at the right temperature to be compacted. Each roller in the rolling train will have its own rolling zone, and this is related to the time available for compaction, as discussed earlier. If the time available changes because of changing weather, the rolling zone should be lengthened or shortened accordingly to maintain uniform density.
- **Rolling pattern** is a generic term used to quantify the types and number of rollers and the specific sequence/order in which they operate for a particular mix type, thickness, and width. Also included is the speed of the rollers, which when using the above variables should be balanced with the other production rates (plant, trucking, and paver). If something changes on the roadway, the rolling pattern will require adjustment. The only constant is the target density, and efforts should be made to make the entire pavement as uniform in density as possible.

In addition, it is important to recognize that there is no one roller type, size, or pattern that will work for every mix in every condition. It is very important to be able to understand the variables and to adapt to them when they change. The goal is to achieve density and preserve the smoothness and texture of the mat.

9.2.2 Need for Compaction

During the compaction of HMA, three very important factors that affect pavement performance take place. The asphalt-coated aggregate particles are pressed together, air voids are reduced, and mix density (weight-to-volume ratio) increases. This squeezing together of the aggregates increases their surface-to-surface contact and inter-particle friction, resulting in higher mix stability and pavement strength.

The reduction of air voids to the optimum level in the mix produces a pavement that is nearly impermeable. In an undercompacted mix, the voids tend to be interconnected, thereby permitting the intrusion of air and water into the pavement structure. Premature pavement distress can then occur. The air oxidizes the asphalt binder, making it brittle and causing the mix to crack under the repeated deflections from traffic loads. Premature aging can also result in premature thermal cracking. Water permeating into and through the pavement can lead to the stripping of the asphalt binder from the aggregate, the weakening of the base and subgrade soils, or freeze-thaw damage in colder climates. Finally, if an optimal level of compaction is not achieved during construction, subsequent traffic will further densify the mix. This will occur principally in the wheelpaths, and the resulting ruts may be a safety hazard for traffic, especially during wet weather.

9.2.3 Mechanics of Compaction

The goal of compacting HMA is to achieve the optimum air-voids content. Behind the paver, the HMA typically has 15 to 20 percent in-place air voids. It is the task of the rollers to reduce that void content to 8 percent or less for dense-graded HMA mixes. Typically, stone matrix asphalt (SMA) mixes should be compacted to 6 percent or less air voids. At these levels, the voids are usually not interconnected and the detrimental effects of air and water (primarily raveling and stripping) are avoided, as illustrated in Figure 9.2. If a mix is compacted to a very low air-voids level (less than 2 percent), it is an indicator that a problem exists in the mix design or mix production itself—for example, too much asphalt binder or dust.

Even with a properly designed mix, compaction of the mat is possible only if two conditions exist: correct mix temperature and mix confinement. At high temperatures, the asphalt binder is in a fluid state, allowing the aggregate particles to readily move past one another and rearrange into a tight mass. As it cools, the asphalt binder stiffens and resists further mix densification. Therefore, compaction must be completed while the mix is hot and workable. Further, no one specific set of temperature ranges applies to all asphalt binders or mixes. For example, mixtures using polymer-modified asphalt binders stiffen at higher temperatures, allowing less time for compaction than mixtures using an unmodified-asphalt binder.

Even when the mat is the correct temperature, the target density can only be obtained when the mix being compacted is adequately confined. This confinement is controlled by three forces: the compressive force of the roller, the support provided by the surface beneath the mat, and the cohesion of the mix. The compressive force of the rollers comes either from their weight or from a combination of weight and dynamic energy due to vibration. The supporting force of the subgrade results from its stability and firmness. The cohesion of the mix is a result of the combined effect of interparticle

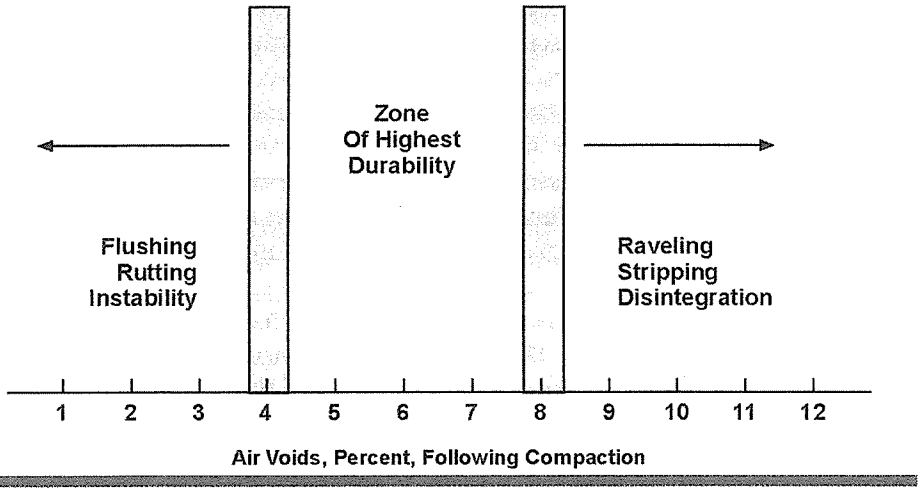


Figure 9.2 Pavement Durability versus Air Voids

friction of aggregates and the asphalt binder's viscosity. When mix density and temperature reach a point where the resisting forces equal the compressive force of the rollers and the resistant forces of the subgrade, equilibrium is reached, as shown in Figure 9.3.

9.2.4 Factors Affecting Compaction

Factors affecting compaction can be categorized into several classes: mix properties, environmental effects, layer (lift) thickness, subgrade and bases, and time available for compaction.

Mix Properties

Properties of various aggregates and asphalt binders have a pronounced effect on the workability of mixes. In addition, the specific mix types chosen for a project may have different target density requirements, which will affect the types of equipment and procedures used. For example, a gap-graded mixture that uses a polymer-modified asphalt binder such as SMA requires a different approach for compaction than does a fine-graded mixture designed for light-duty traffic. These properties and the temperature of the mix at the time of compaction must be considered when deciding on a compaction procedure.

Aggregate

Gradation, surface texture, angularity, and characteristics of the filler/dust affect the workability of the mix. As the maximum aggregate size or the percentage of coarse aggregate in the mix increases, the workability decreases and greater compactive effort is required to achieve target density.

Similarly, mixtures using aggregates with a rough surface texture require greater compactive effort, as opposed to a smooth-aggregate surface. Many

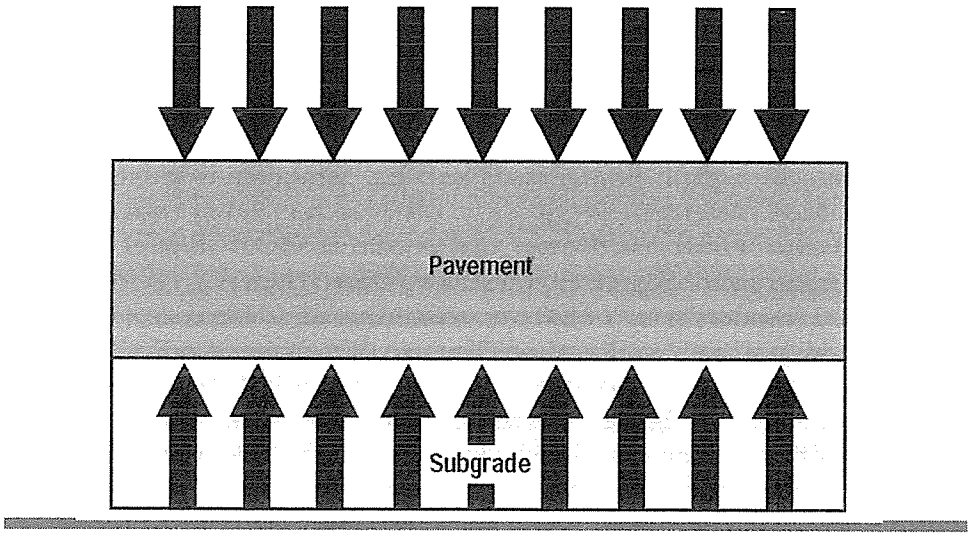


Figure 9.3 Forces at Work During Compaction

mixtures used today on high-volume and/or heavy-load facilities, such as interstate highways and major airports, have specifications that require high percentages of crushed aggregate. Mixtures designed to resist rutting under heavy traffic often require more compactive effort applied at higher temperatures than those designed for light-duty traffic.

Mixtures produced from gravels typically have more rounded particles than quarried stone and are thus more workable and easier to compact. Natural sands, which commonly have rounded particle shapes, are often added to lower-traffic volume mixes in the interests of economy and to improve workability. Too much sand, particularly in the middle particle sizes—around the No. 30 (0.60 millimeter) mesh sieve—will result in tender mixes (mixes with high workability but low stability). Tender mixes are easily displaced by heavy rollers and too much rolling. They are often susceptible to scuffing and shear deformation by traffic after several weeks in place.

The fines or filler content in the mix will also affect the compaction process. The combination of filler and asphalt binder is what glues the aggregates together in HMA pavements; therefore, the mix should contain sufficient fines to combine with the asphalt binder to produce the necessary cohesion when the mix cools. The addition of mineral filler will help offset the tenderness or slow-setting properties of mixes containing too much sand. Conversely, if a mix contains too many fines it will become "gummy" and very difficult to compact.

Asphalt Binder

At room temperatures, asphalt binder is virtually a solid, whereas at 250°–300°F (121°–149°C), it is a fluid. As previously mentioned, for a mix to be properly compacted, the asphalt binder must be fluid enough to permit the aggregate particles to move past one another. In effect then, the asphalt binder acts as a lubricant during compaction. As the mix cools, the asphalt binder solidifies and resists compaction. In general, once mixtures made with an unmodified asphalt binder fall below 175°F (80°C), the asphalt binder, in combination with the fines in the mix, begins to bind the aggregate particles firmly in place. Consequently, productive compaction of the mix without damaging it is extremely difficult once the mix has cooled below that point.

The grade of asphalt binder used and the temperature at which the mix is produced determine its viscosity. Other factors being equal, higher-viscosity asphalt binders in the mix may require a slightly higher compaction temperature and/or greater compactive effort. Generally, Performance Graded (PG) binders with a high temperature grade difference of 92°C or more require polymer modification. The use of polymer-modified asphalt binders in HMA has increased steadily since the mid-1990s. These binders must be heated to higher temperatures to achieve the same fluidity as unmodified binders for mixing and compaction, and compaction must be completed more quickly—before the mixture temperature drops below 240°F (115°C). Care must be taken not to overheat any asphalt binder, or damage may occur. Consult the binder manufacturer for recommended temperatures and handling practices.

The effective asphalt binder content of the mix also affects workability. As the asphalt content increases, the film thickness of the asphalt binder on the aggregate particles increases. At compaction temperatures, this increased film thickness increases the lubricating effect of the asphalt binder and up to a certain point makes compaction easier.

Mix Types

Three main types of mixes are used in HMA construction: dense-graded, open-graded, and SMA. These different mix types are discussed in Chapter 4, "Hot Mix Asphalt Design." Each has specific characteristics that will affect its ability to be compacted. In general, the more compactive effort required in the laboratory for the mix to achieve target air-voids requirements, the tougher it will be to compact in the field. Open-graded mixtures generally do not require a great amount of compactive effort in the field.

Mix Temperature

The temperature at which an HMA mixture is produced affects both the ease of compaction and the time available for compaction. The time available for compaction increases with mix temperature, but there are limits as to how high the production temperature should be to avoid damaging the asphalt binder. Upper limits for mix temperature are discussed in Chapter 2, "Petroleum Asphalts," and more specifically in Asphalt Institute's (EC-101), *Best Management Practices to Minimize Emissions During HMA Construction*, which covers best practices for storage and handling of asphalt binders. The upper temperature limits increase with higher PG grades and are generally higher for modified binders. The best temperature to begin rolling is the maximum temperature at which the mix will support the roller without moving horizontally.

At the time of placement, the mix temperature is essentially uniform throughout the thickness of the lift. However, the top and bottom surfaces cool more rapidly than the interior portion of the mat. Extreme temperature differences can lead to "heat checking" or other surface deficiencies, or to honeycombing near the bottom of a thick lift placed during cold weather. Careful surface inspection during compaction and observation of the condition of core specimens removed from the pavement is necessary to judge whether these problems exist.

Environmental Effects

The rate at which the mix cools affects the length of time during which density can and must be achieved. The following conditions all affect the rate of cooling of the HMA: ambient (air) temperature, the temperature of the surface upon which the mix is being placed, the wind speed, and the time of day. Cool air temperatures, strong winds, and cool surfaces all shorten the time for compaction. This may make compaction more difficult. Paving during the day will usually allow for a longer paving time because of the sun's effect on the air temperature and on the underlying surface temperature. Conversely, night paving tends to allow less time for compaction.

Layer (Lift) Thickness

Generally, it is easier to achieve target density in thicker lifts of HMA than in thin ones. The thicker the lift, the longer it retains its heat and the longer the time for achieving compaction. This is beneficial when rolling lifts of highly stable mixes that are difficult to compact, or when paving in cool weather.

A general rule of thumb for mixes above the maximum density line (see Chapter 4, "Hot Mix Asphalt Design") would be to use at least three times the nominal maximum aggregate size for the minimum lift thickness. For example, a ½-inch (12.5 millimeters) fine-graded mix would have a minimum compacted lift thickness of 1½ inches (37.5 millimeters). A coarse-graded mix (having a gradation that passes below the maximum density line) requires a 4-to-1 rule or minimum lift thickness of four times the nominal maximum aggregate size. For example, a ½-inch (12.5 millimeters) coarse-graded mix would have a minimum compacted lift thickness of 2 inches (50 millimeters). The reason for the different criteria is to provide sufficient room for the particles to realign and pack during compaction.

Increased layer thickness can permit lower mix temperatures to be used because of the reduced rate of cooling. For example, paving thick layers in the middle of the summer may lead to an excessively long wait for the mix to cool before the finish rolling can be completed. Additional rolling during this time may actually damage the mat.

Uniform lift thickness is important as well. In variable thickness layers (such as those used to correct cross-slope or to level the profile), it is best to use a prescribed rolling pattern rather than a density target, to ensure that the best possible uniform density is achieved. Use of a pneumatic-tire roller on uneven surfaces will also help. If the thickness of the layer is less than two and a half to three times the nominal aggregate size, it may be impossible to achieve the density requirements in the specifications.

Subgrade and Bases

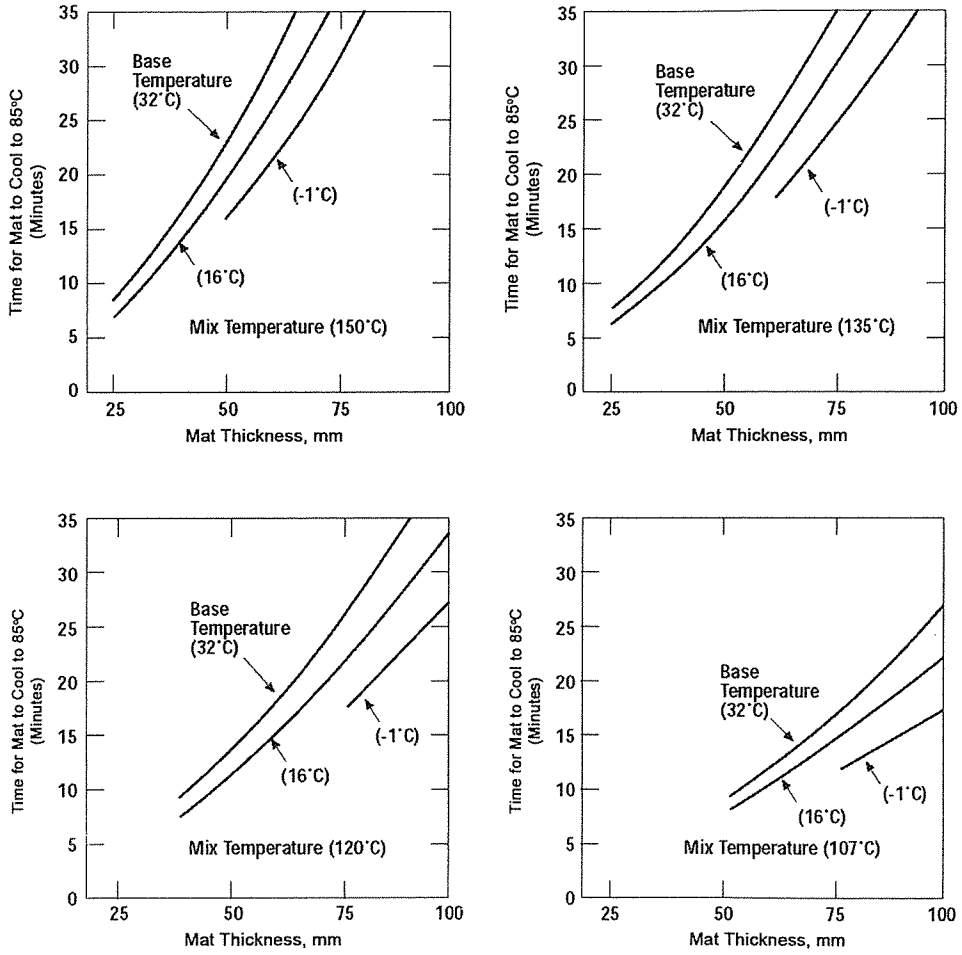
The condition of the subgrade and/or base that is being paved upon will have a significant impact on the final density achieved. The stiffness of the underlying layer plays an important part in helping to provide the confinement necessary for compaction. Attempting to bridge a soft base or subgrade with HMA mix is not recommended. In addition, the temperature of the base and the moisture condition can either help or hurt the compaction process. Cold, wet bases tend to pull heat out of the HMA mix, causing it to cool rapidly, thus making compaction more difficult.

The base and subgrade layers must be level and true to line and grade so that the HMA layers are uniform in thickness and section. As previously mentioned, a soft or weak base or subgrade will create difficulties in obtaining the desired uniform density.

Time Available for Compaction

All these factors (mix temperature, air temperature, wind, time of day, mat thickness, and base condition and temperature) will affect the rate of cooling and thus the time available for compaction. Several computer

programs are available for calculating the time available for compaction based on a set of site-specific and user-defined variables. These programs are helpful tools for understanding the effects of different variables on compaction. Of all the variables, layer thickness appears to have the greatest effect on the time available for compaction. Thin lifts paved at night under cool conditions can result in compaction times as brief as 6 minutes, as shown in Figure 9.4. Under these conditions, consistently achieving the target in-place air-voids level is exceedingly difficult.



Wind velocity - 10 knots. Atmospheric temperature - same as base.

Note: "Base Temperature" is the temperature of the surface upon which the asphalt mat is placed.

85°C is the temperature of the mat measured 6 to 12 mm below the mat surface. The average temperature of the entire mat thickness when this temperature is reached is approximately 80°C.

Placing thicknesses less than those shown by the curves is not recommended in subgrades of -1°C (base temperature).

(Conversion: 25.4 mm = 1 in. °F = [(9/5)°C] + 32)

Figure 9.4 Time Allowed for Compaction, Based on Temperature and Thickness of Mat and Temperature of Underlying Base

9.2.5 Cool Weather Paving

Placing and compacting HMA is difficult in cool weather and poses more challenges than just the reduced time available for compaction. It is particularly important to ensure that the equipment is properly maintained and that the sequence of delivery, placement, and rolling is well coordinated. Special attention should be given to pneumatic-tire rollers to keep the tires warm to avoid pickup, which will be discussed later.

Also in cool weather, the temperature differential between the air and the mix is substantial, causing the surface to cool much more rapidly than the internal portion of the mix. This differential cooling may cause a thin crust to form at the surface. This crust might develop microcracks (called checking) that later may cause the pavement surface to ravel prematurely. Cold surfaces may also cause immediate solidification of the mixture that is in contact with the surface, resulting in "honeycombing" in thick mats and possibly poor adhesion or slippage in thin mats.

In summary, materials, mix types, and environmental conditions can have a profound impact on the compatibility of the mix. These effects are summarized in Table 9.1.

Table 9.1
Summary of Influences on Compaction

Item	Effect	Corrections*
Aggregate		
• Smooth-surfaced	Low interparticle friction	Use light rollers Use lower mix temperature
• Rough-surfaced	High interparticle friction	Use heavier rollers
• Unsound	Breaks under steel-wheel rollers	Use sound aggregate Use pneumatic rollers
• Absorptive	Dries mix—difficult to compact	Increase asphalt-binder content in mix
Asphalt Binder		
• Viscosity—high	Particle movement restricted	Use heavier rollers Increase mix temperature
• Viscosity—low	Particles—move easily during compaction	Use light rollers Decrease mix temperature
• Quantity—high	Unstable and plastic mixes	Decrease content in mix
• Quantity—low	Reduced lubrication—difficult compaction	Increase content in mix Use heavy rollers
Mix		
• Excess coarse aggregate	Harsh mix—difficult to compact	Use heavy rollers
• Oversanded	Too workable—difficult to compact	Reduce sand in mix Use light rollers
• Too much filler	Stiffens mix—difficult to compact	Reduce filler in mix Use heavier rollers
• Too little filler	Low cohesion—mix may come apart	Increase filler in mix
Mix Temperature		
• High	Difficult to compact—mix lacks cohesion	Decrease mixing temperature
• Low	Difficult to compact—mix too stiff	Increase mixing temperature
Layer Thickness		
• Thick lifts	Hold heat—more time to compact	Roll normally
• Thin lifts	Lose heat—less time to compact	Roll before mix cools Increase mix temperature Shorten rolling zone
Weather Conditions		
• Low air temperature	Cools mix rapidly	Roll before mix cools Increase mix temperature Shorten rolling zone
• Low surface temperature	Cools mix rapidly	Roll before mix cools Increase mix temperature Shorten rolling zone
• Wind	Cools mix—crusts surface	Increase lift thickness

*Corrections may be made on a trial basis at the plant or job site. Additional remedies may be derived from changes in mix design.

9.3 Compaction Equipment

Compaction is performed using any of several types of rollers that apply force (either static or dynamic) to the mat, reorienting aggregate particles to reduce the volume occupied by the mixture.

Self-propelled rollers are by far the most common type of compaction equipment used for HMA construction. Towed-type rollers should not be used for HMA compaction, but hand-operated walk-behind rollers, vibrating plates, or even hand tamps are often used in small areas inaccessible to larger rollers. Typical self-propelled compaction rollers include these three types:

- Static steel-wheel
- Pneumatic-tire
- Vibratory

9.3.1 Static Steel-Wheel Rollers

Steel-wheel rollers by configuration are two-axle tandem rollers or three-wheel rollers, as shown in Figure 9.5. Usually, the drums are solid across the width, though models with split drums are available. The split drum allows the rotation speed of the drum to change (from one side to the other) when turning, minimizing shear stresses on the surface of the

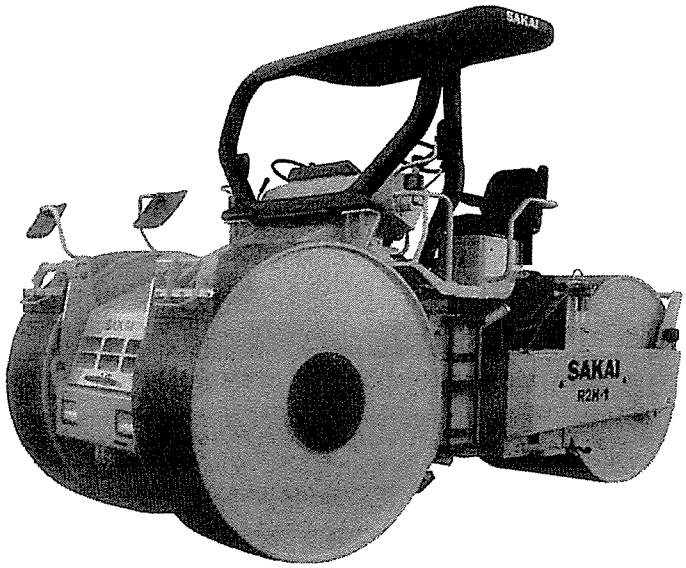


Figure 9.5 Static Steel-Wheel Roller

lift that may cause tearing or cracking. Steel-wheel rollers used for rolling HMA include a water-spray system and scrapers to moisten the drums to help prevent mix from sticking to the drums.

Steel-wheel rollers should be checked for wear on the surface of the drum. A sharp metal straightedge may be used for this check. The roller should not be used if grooves or pits have worn into the rolling drum. Scrapers should be routinely inspected and replaced if they are excessively worn.

Weight

Steel-wheel rollers can vary in weight from 3 to 14 tons (2.7 to 12.7 tonnes) or more. Most can be ballasted to increase the weight. Roller widths range from 40 inches to 54 inches (102 centimeters to 137 centimeters). A typical steel-wheel roller would have an unballasted weight of 8 tons (7.9 tonnes) and a ballasted weight of 10 tons (9.8 tonnes) with 50-inch (127 centimeters) wide drums. Wider and heavier rollers are available from a variety of manufacturers.

Pounds per Linear Inch

Because steel-wheel rollers vary in width and weight, a simple means was developed to help quantify the compactive effort applied by static rollers. By dividing the weight of the roller by the width of both drums, the overall static linear load, expressed in pounds per linear inch (PLI) or kilograms per millimeter of roller, can be determined. For example, an 8-ton roller with a 50-inch drum width would have an overall PLI of 160 PLI. The PLI can be used to match up rollers of different sizes and manufacturers when establishing a rolling pattern.

Adjusting Compaction Force

The two ways to adjust the compactive force of a steel-wheel roller are to adjust the ballast (weight) of the roller and the speed. Adding weight will increase the static linear load and apply greater force on the surface with each pass or coverage the roller makes. Adjusting the speed of the roller affects the dwell time that the drum has on the surface of the HMA. Speeding up a roller decreases the dwell time and lowers density.

9.3.2 Pneumatic-Tire Rollers

Pneumatic-tire rollers are also called rubber-tire rollers, or seal rollers. Self-propelled pneumatic-tire rollers have two to eight wheels in front and four to eight wheels in the rear, though most will fall into the seven- or nine-total-wheels classes. The wheels on these rollers generally oscillate (axle moves up and down). An example of a pneumatic roller is shown in Figure 9.6.

Unlike a steel-wheel roller, the surface of a pneumatic-tire roller conforms to the shape of the underlying surface. This allows the roller to contact all points on the surface and also to manipulate particles differently than a steel-wheel roller.



Figure 9.6 Self-Propelled Pneumatic-Tire Roller

Pneumatic-tire rollers have several advantages:

- They provide a more uniform degree of compaction than steel-wheel rollers.
- They improve the seal near the surface, thus decreasing the permeability of the layer.
- They orient the aggregate particles for greatest stability, as high-pressure truck tires do after using the HMA surface for some time.

The main disadvantage of a pneumatic-tire roller is the tendency for the tires to pick up fines from the surface of the HMA mix—unless the tires are kept clean and warm. This can be a serious problem with mixtures using polymer-modified asphalt binder and especially for SMA. Although pneumatic-tire rollers have been used successfully to compact SMA, their use is discouraged with this mixture classification.

Several self-propelled pneumatic-tire rollers have a device to change tire inflation while the roller is operating. This inflation-on-the-run system automatically maintains any preset tire pressure or can raise or lower the inflation pressure while the roller is operating. Some compacting conditions and requirements require different inflation pressures. However, tire inflation pressures should all be very close between tires and checked at regular intervals.

Weight

Self-propelled pneumatic-tire rollers vary in weight from 10.5 to 30 tons (9.5 to 27.2 tonnes). Ballast can be added to the machines to increase the weight. Widths of these rollers range from 60 to 90 inches (524 to 2286 millimeters).

Contact Pressure

The contact pressure of a pneumatic-tire roller is directly related to the wheel load divided by the contact area (footprint) that the tire has in contact with the pavement surface. The ply rating of the tire determines the maximum and minimum inflation pressures, thereby affecting the contact area and resulting contact pressure. Running the tires at the mid to lower end of the inflation pressure range for the particular tire will help improve the surface texture by sealing the surface. Running the tire inflation pressure higher increases the compactive effort of the roller. All the tires should have the same ply rating and the same inflation pressure.

Adjusting Compactive Force

There are several ways to adjust the compactive force of a pneumatic-tire roller. The first is to add ballast, which will increase the mass of the roller. The second means of adjusting the compactive force is to increase the air pressure in each of the tires per the tire inflation pressure recommendations. Increasing air pressure reduces the contact area (footprint) and increases the contact pressure of the roller. The third means of adjusting the compactive effort is to adjust the speed of the roller (just like the steel-wheel roller). Adjusting the speed of the roller affects the dwell time the tires have on the surface of the HMA.

Tire Pickup

When using pneumatic rollers, keeping the tires hot is the most effective means of preventing pickup. Running the roller back and forth on a previously placed mat for at least 10 minutes will help warm the tires; once paving has started, the roller can slowly work into the hot mix. If needed, applying a small amount of nonfoaming detergent or water soluble release agent on the wetting mat of a pneumatic-tire roller at the beginning of rolling operations helps prevent the HMA from sticking to the tires before they warm up. Do not use diesel or other petroleum fuels/solvents as a release agent. Keeping pads and scrapers on each tire will prevent pickup from getting worse if a tire does pick up in an area. If tires start to pick up, move the roller off the mat, clean the tires carefully, then start again. Skirting around tires helps minimize heat loss due to wind.

9.3.3 Vibratory Rollers

Vibratory rollers are the most versatile and effective rollers for compacting HMA. They may include one or two smooth-surface steel drums, as shown in Figure 9.7, that have eccentric weights spinning around a shaft within the drum. Vibratory rollers can be used for compacting any type of HMA mixture but generally are not used in the vibratory mode when the mat thickness is 1.5 inches (37.5 millimeters) or less to avoid fracturing the aggregate and damaging the mixture. The compactive effort of vibratory rollers can be adjusted for the mix type and base conditions on the project by varying the frequency and amplitude. Spilt-drum vibratory rollers can also



Figure 9.7 Vibratory Roller

be operated in "mixed mode," whereby only one drum is operated in vibratory mode. As with static steel-wheel rollers, vibratory rollers should be checked for wear on the surface of the drum. A sharp metal straight-edge may be used for this check. The roller should not be used if grooves or pits have worn into the rolling drum. Scrapers should be routinely inspected and replaced if they are excessively worn.

Weight

Vibratory rollers vary in static weight from 2.5 to 18 tons (2.3 to 16.4 tonnes). Widths vary from 39.4 to 84 inches (1000 to 2134 millimeters). The static linear load is generally less for vibratory rollers than for static steel-wheel rollers.

Frequency

Vibratory rollers may be used in static mode while compacting thinner lifts of HMA, but the use of dynamic force is what makes them so effective when compared with static rollers. Frequency is expressed in vibrations or impacts per minute, or hertz (Hz), and is a measure of the rotational speed of the eccentric weight inside the drum of the roller. Typical frequencies of rollers used on an HMA pavement range from 2,500 to over 4,000 vibrations per minute (vpm) (42–67 hertz). The high-frequency drum movement puts the aggregate particles in motion, allowing them to slide past one another more easily under the weight of the drum. Mixes with high percentages of aggregate with crushed faces benefit the most with vibratory rollers. The general rule of thumb is to use highest-frequency setting available. High frequency also allows the roller to be operated at a greater forward

speed, increasing the efficiency of the roller. The frequency of each drum should be routinely checked using a reed tachometer.

Amplitude

Simply explained, *amplitude* is the up-and-down motion of the drum that is caused as the eccentric weight spins inside the drum. The higher the amplitude, the greater the vertical movement generated, and the greater the force exerted on the HMA mix. The general approach is to use the lowest-amplitude setting that will achieve density. However, amplitude is tied closely to the thickness of the layer being placed. Higher-amplitude settings are appropriate when lift thicknesses are 3 inches (75 millimeters) or greater. Thicker layers can handle a higher amplitude. Care must be taken not to roll too much with a vibratory roller because it is possible to overcompact. Too much force can damage the HMA surface, crushing aggregate particles.

Impact Spacing

The vibratory roller should always be moving when the vibration system is enabled. As the drum vibrates and the roller is moving, the drum is lifting and dropping on the mat, creating small impacts in a uniform pattern. This pattern is difficult to see except at night when the light angles are low. The general rule of thumb for impact spacing is 10–14 impacts per foot (33–46 impacts per meter). This allows the entire surface to be impacted as the roller travels in vibratory mode. A number of manufacturers have indicators at the operator's station to indicate whether the roller is being operated properly. If the frequency of the roller is known, it can be divided by the desired impact spacing to establish the speed of the roller. For example, if a roller can develop 3500 vibrations per minute and a 10 impact-per-foot impact spacing is desired, the optimum speed would be 3500/10 = 350 feet per minute (107 meters per minute). In summary, impact spacing, vibration, and amplitude can be used to adjust the compactive effort of a vibratory roller. Slowing the forward speed of the roller decreases the impact spacing, causing the number of impacts per foot to increase as shown in Figure 9.8.

9.3.4 Guidelines for Choosing a Roller

Any type or combination of rollers can be used to compact HMA mixes. The key to roller selection is efficiency and effectiveness. A general rule of thumb is that lighter rollers are better suited to more tender mixes placed in thin layers, and heavier rollers are better suited to thicker, stiffer layers. A light roller on an interstate highway would be neither efficient nor effective. Wider rollers can cover the mat quickly and are more efficient and effective on mainline highways, but their use in a parking lot with many turns and slope changes would be ineffective.

Thin Lifts (<1.5 inches compacted)

On thin lifts, most rolling operations can be completed with static steel and/or vibratory rollers operating in the static mode. A rolling pattern

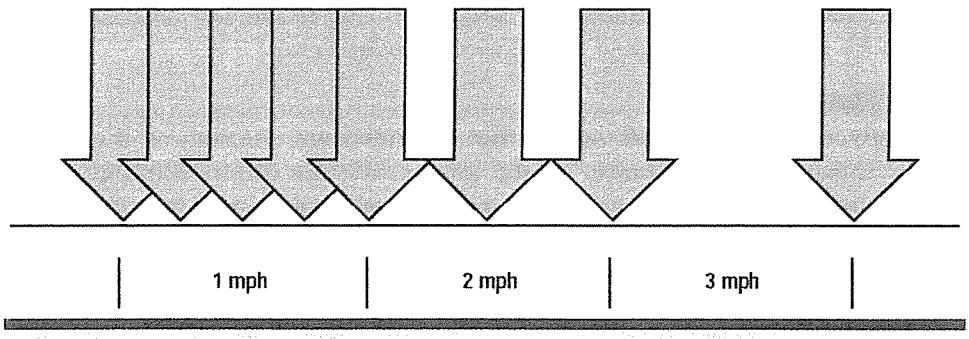


Figure 9.8 Impact Spacing

using several coverages is usually adequate for achieving the desired density. Alternately, some have used pneumatic-tire rollers as the primary compaction roller for thin HMA overlays. This is particularly true for thin leveling courses because the pneumatic tires conform to the shape of the surface and will not bridge over uneven areas.

Thicker Lifts

Although thicker lifts allow more time to achieve density, the greater mass of material often requires more compactive effort. Vibratory rollers compact more efficiently than static rollers and are almost always used during compaction of thick lifts. A typical rolling train consists of vibratory rollers operating as breakdown rollers, followed by vibratory or pneumatic intermediate rollers, and finishing with a static steel-wheel roller or vibratory roller operating in the static mode.

Leveling Courses—Uneven Bases

Pneumatic-tire rollers are very useful when compacting leveling courses over uneven surfaces or when placing variable-depth layers. The action of the tires allows for compaction into the uneven areas that a wide steel-drum might bridge over.

Non-Mainline Applications

Most rolling patterns are described to be set up for constant-width paving or mainline paving. Often paving is done in irregular areas, curved sections, cul-de-sac, or parking areas. Static rollers are common in these areas, as are pneumatic and vibratory rollers. Again, a combination of rollers generally works best. Split-drum steel rollers can make tighter turns without tearing or cutting the mat.

Statistics-Based Specifications

These types of specifications are based on how uniform and how close to the target all the density tests are. Monitoring density is vital, and adjusting the roller pattern as conditions change will ensure the best possible specification with a statistics-based specification such as Percent Within Limits (PWL).

9.3.5 New Technology

Perhaps the most significant development in recent years has been the increase in frequency available for "highway class" vibratory rollers. Since 2000, most roller manufacturers have developed large tandem vibratory rollers capable of operating at frequencies of over 3000 vibrations per minute (50 hertz); some can operate at 4000 vibrations per minute (67 hertz) or higher. This permits a single roller to cover a much larger area while maintaining the minimum recommended impact spacing.

There have been substantial advances in equipment technology in the compaction area, and this trend appears to be growing. Advances in sensory and vibratory systems have resulted in the development of rollers capable of automatically adjusting the amplitude based on the stiffness of the HMA pavement. As the mix becomes denser (also stiffer) through the compaction process, several manufacturers have equipment that adjusts the intensity and direction of the amplitude to maximize efficiency. These concepts and equipment were first developed for soil and embankment compaction, and are often called Intelligent Compaction (IC).

Rollers may have built-in temperature sensors to help the operator better understand the mix as it cools. Technology exists to monitor and record the stiffness of the pavement in relation to the location of the roller and to provide printouts or color displays of areas needing additional coverage. Advances in pneumatic-tire technology have yielded tires that HMA doesn't stick to and stiffer tires that help focus the energy into the compaction process.

Vibratory pneumatic rollers have been developed to harness the advantages of both types of vibration and pneumatic-tire compaction technology. Continuing advances in compaction equipment seem certain.

9.3.6 General Operational Notes

Before any roller is used on a project, it should be inspected to see that it is in good mechanical condition. Where applicable, the following items should be checked:

- Mechanical condition, including verification of frequency for vibratory drums, using a reed tachometer
- Water-spray systems, pads, scrapers operational
- Total weight of roller
- Weight per inch of width (steel-wheel rollers)
- Tire inflation pressure and average ground contact pressure (pneumatic-tire rollers)
- Precise steering

9.4 Rolling Procedures

The goal of the compaction process is to achieve the required density in as few coverages as necessary while preserving the smoothness and texture created by the paver. Rolling should start as soon as possible after the HMA has been spread and can support the weight of the rollers without lateral distortion.

9.4.1 Test Strip

A test strip should be built at the beginning of every project for each mix or thickness to determine the proper roller pattern. The test strip should be built at the same paving speed at which normal operation is anticipated. The in-place density is checked after each coverage, using a nondestructive gauge (nuclear or non-nuclear). A relationship between number of coverages and density is determined while monitoring the temperature of the mixture. This process helps determine the optimum number of coverages for each roller in the rolling train and estimate the dimensions of rolling zones for each roller. As shown in Figure 9.9, density increases with each coverage, then peaks. The peak is relatively flat with static steel-wheel rollers but very sharp with vibratory rollers. Care must be taken to not go beyond the peak with vibratory rollers because the density will decrease due to the mix being damaged.

Subsequently, during the compaction process, those involved must be particularly observant and knowledgeable. A qualified quality-control technician using a calibrated density gauge is invaluable in determining the appropriate rolling pattern and adjusting the pattern as needed to ensure that the target density level is achieved.

9.4.2 Determining Roller Requirements

The width of paving should be considered when selecting rollers. If a wide pavement layer is constructed using narrow rollers, the rollers will need to make many passes to completely cover the surface. This takes time during which the mix will cool and differential density can result. Generally, the width of the pavement is no more than three times the width of the roller. Often the contractor selects two wide rollers operating slightly offset from each other so that they effectively cover the width of the mat each time they make a pass over the mat.

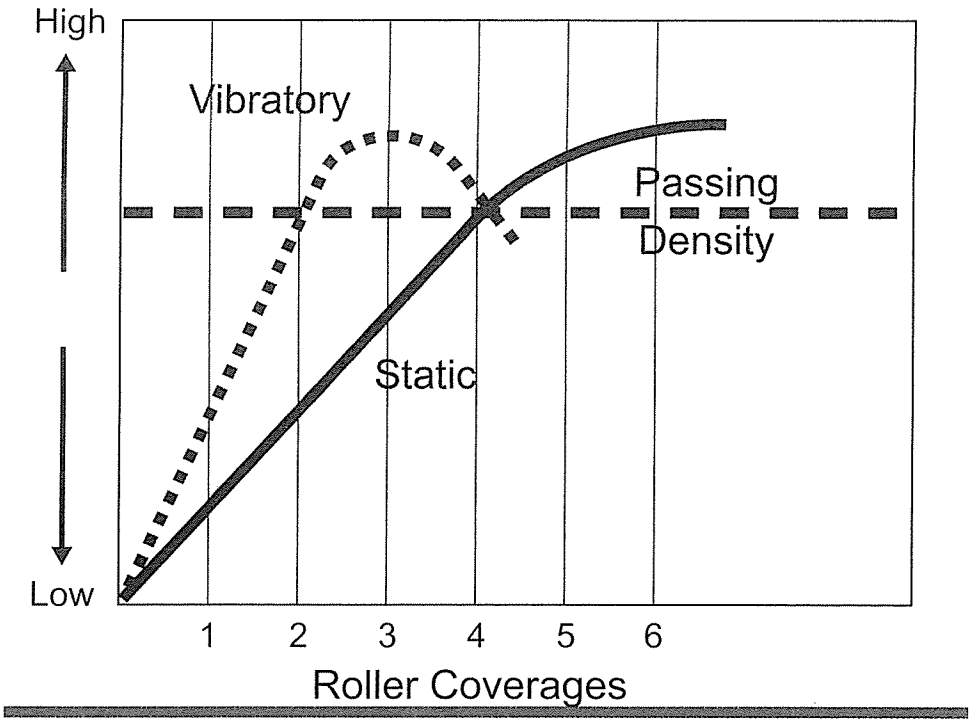


Figure 9.9 Density Growth Curve

9.4.3 Lift Thickness

The thickness of the lift being placed will have an effect on the time available for compaction and on the types of equipment used to compact the HMA pavement. If the lift is less than 1.5 inches (30 millimeters) after rolldown (compaction), it can be compacted with combinations of static steel-wheel rollers, vibratory rollers operating in static mode, and/or pneumatic rollers. Thin layers tend to cool quickly, requiring attention to the compaction process so that the desired level of density is achieved. Thicker lifts of HMA tend to hold heat longer, allowing more time for compaction, and may require more effort to compact due to the thickness. More rollers may be needed, including vibratory rollers operating at higher amplitude settings, and combinations of vibratory, pneumatic, and static rollers.

9.4.4 General Rolling Hints

Two or more rollers are needed on most projects, other than small jobs such as driveways. During rolling, roller wheels are kept moist with only enough water to avoid picking up material. Do not use petroleum-based fuels or oils on roller drums or pneumatic tires. Make sure that the water tanks are full and that the water-spray system is operating properly, with the appropriate pads and scrapers in place. Rollers should move at uniform

speed with the drive roll or wheels nearest the direction of travel. The speed should not exceed 3 miles per hour (5 kilometers per hour) for steel-wheel static rollers or 5 miles per hour (8 kilometers per hour) for pneumatic-tire rollers. As mentioned earlier, the frequency and desired impact spacing can be used to establish the speed of a vibratory roller. Higher-frequency (4000 vibrations per minute) vibratory rollers can provide 10 impacts per foot at over 4.5 miles per hour (7.3 kilometers per hour).

Rollers must be kept in good condition, capable of reversing direction without jerking or backlash. The line of rolling should not be suddenly changed nor should the direction of rolling be suddenly reversed, thereby displacing the mix. Any pronounced change in direction should be made on stable material. Care should be taken when turning wide rollers to prevent lateral shoving of the mix, especially near unsupported edges.

If rolling causes material displacement, the affected areas should be loosened at once with lutes or rakes and restored to their original grade with loose material before being rerolled. Rollers should not be permitted to stand on the finished surface before it has thoroughly cooled or set.

9.4.5 Sequence of Roller Operations

The number of rollers required on a project is related to the type of mix, the production rates, and the desired density levels. It is impossible to establish one roller pattern that will achieve density on all mixes. However, the generally accepted sequence has three phases: breakdown or initial rolling, intermediate rolling, and finish rolling. Breakdown rolling provides most of the increase in density during the compaction operation. Intermediate rolling further densifies and seals the surface. Finish rolling removes roller marks and other blemishes left from previous rolling and minimally increases density.

Breakdown Rolling

Breakdown rolling is usually accomplished with steel-wheel rollers. Vibratory or static-weight tandem rollers are recommended. Pneumatic-tire rollers, however, have also been used successfully for breakdown rolling, but normally should be limited to base courses.

The weight of the roller used for breakdown rolling depends to a large degree on the temperature, thickness, and stability of the mix being placed. Generally, a roller weighing 8–12 tonnes (7300–10,900 kilograms) is used for this operation.

In general, it's best to start the rolling operation on the low side of the mat, which is usually the outside of the lane being paved, and progress toward the high side. The reason is that these HMA mixtures, when hot, are not as stable as current mix types and tend to migrate toward the low side of the mat under the action of the roller. If rolling is started on the high side, this migration is much more pronounced than if rolling progresses from the low side. When adjoining lanes are placed, the same

rolling procedure should be followed. This assures that proper cross-slope and/or superelevations are maintained, but only after compaction of fresh mix at the longitudinal joint with 6–8 inches (150–200 millimeters) of roller width.

A rolling pattern that provides the most uniform coverage of the lane being paved should be used. Rollers vary in widths, and a single recommended pattern that applies to all rollers is impractical. For this reason, the best rolling pattern for each roller being used should be developed and followed to obtain the most uniform density across the lane.

The rolling pattern includes not only the number of passes but also the location of the first pass, the sequence of succeeding passes, and the overlapping between passes. Rolling speed should be consistent, with actual speed depending on the roller type, as discussed earlier. In addition, sharp turns and quick starts or stops are to be avoided because they can leave tears or bumps in the mat.

Vibratory rollers allow for more options than static rollers. Dual-drum vibratory rollers have power to both drums, so direction is not an issue. It is possible to selectively vibrate the front drum, the rear drum, or both. It is also possible to vibrate in one direction and run static in another. All these variations not only offer limitless possibilities but also increase the chance that a complicated rolling pattern will be forgotten or ignored. The best rolling pattern is the one that results in a consistent density that meets the target values. A simple pattern will be more consistently followed than a complicated one.

For thin lifts, a recommended rolling pattern for static steel-wheel rollers is shown in Figure 9.10. The rolling operation should start from the edge of the mat on the low side with the roller moving forward as close behind the paver as possible. The second movement of the roller should be to reverse in the same path until the roller has reached previously compacted material. At this point, it should swing over and move forward along the third path, again following as close as possible behind the paver. The fourth movement is reversal in the third path and a repetition of the previous operation. After the entire width of the mix being placed

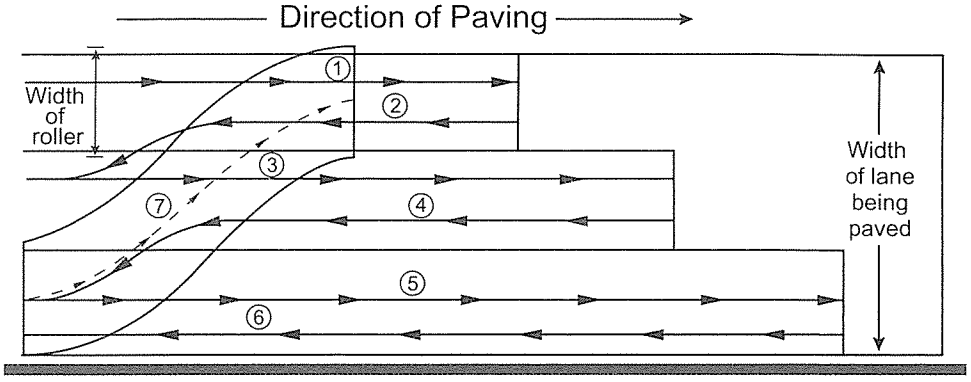


Figure 9.10 Correct Rolling Pattern

has been covered in this fashion, the roller should swing across the mat to the low side and repeat the process. With this pattern, the lap of the roller with succeeding passes need not be more than 3–4 inches (75–100 millimeters). The roller should not turn in a manner that causes it to stop perpendicular to the direction of the lane being rolled.

With static steel-wheel rollers, the operation should progress with the drive wheel forward in the direction of paving. If the breakdown pass of the roller is made with the tiller wheel forward, the pushing force and the weight is slightly ahead of vertical, causing material to push up in a wave in front of the wheel. The greater weight of the drive wheel carries out the compaction, while the turning force tends to tuck material under the front of the wheel as shown in Figure 9.11.

There are exceptions to rolling with the drive wheel forward, however. They usually occur when superelevations are being constructed or on steep grades (more than 5 percent). The exceptions occur when, due to these high grades, the drive wheel of the roller begins to chatter on the

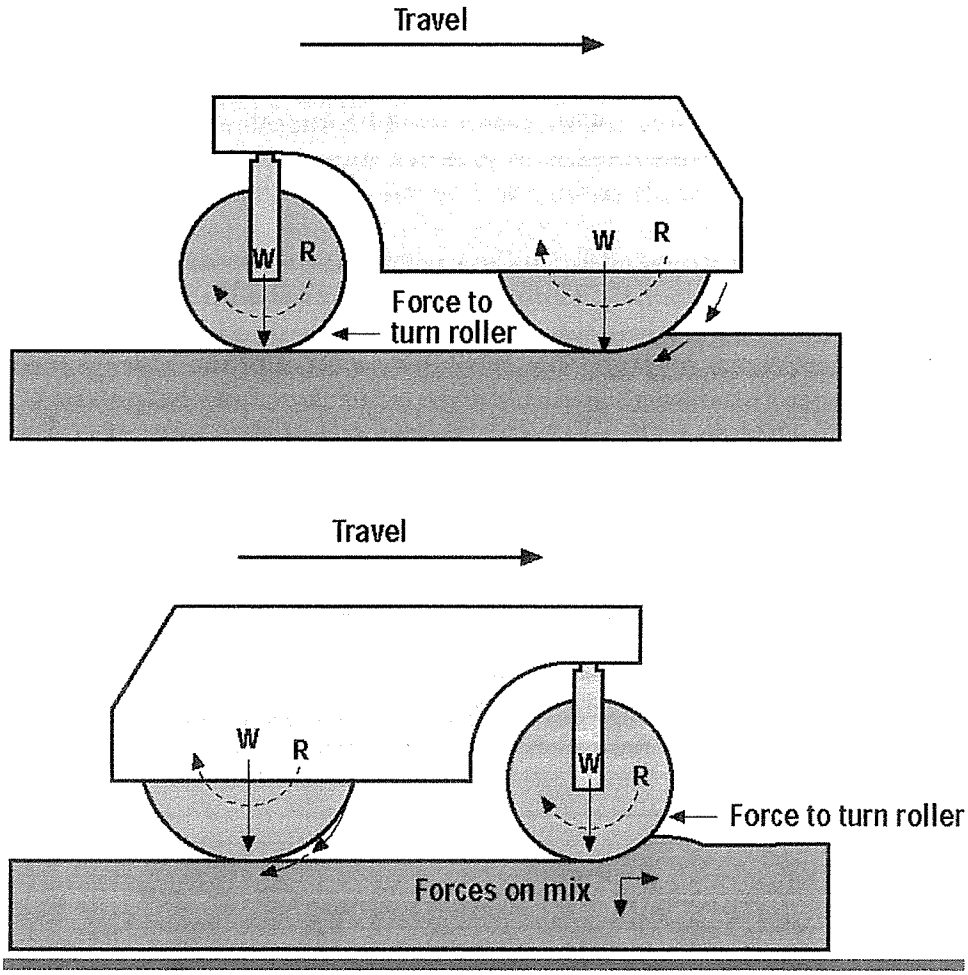


Figure 9.11 Forces Acting when Tiller Wheel or Drive Wheel is Forward

mat, causing displacement of the mix and a very rough surface. In these cases, the roller must be turned around to allow the tiller wheel to partially compact the material so that the drive wheel can then proceed over it. On extreme superelevations such as that of racetracks, the roller may need to be equipped with outriggers to keep it from overturning.

Water-spray systems should be used to keep mix from sticking to the drums. Many rollers have intermittent spray systems to avoid using an excessive amount of water, which can cool the surface of the mat and the face of the drum. It is also common to use liquid detergent or a water-soluble release agent in the water to help avoid pickup.

The best practice is to have a quality-control technician present with a calibrated density gauge and infrared thermometer to monitor temperature and density and to adjust the rolling pattern as needed to ensure the density level is achieved. Having a reed tachometer available to confirm the frequency of vibrating drums is also beneficial for setting the proper forward speed of the rollers to maintain the proper impact spacing.

Intermediate Rolling

Secondary or intermediate rolling should closely follow breakdown rolling while the HMA mix is still hot and compactable.

Pneumatic-tire or vibratory tandem rollers may be used for intermediate rolling. Tire contact pressures should be as high as possible without causing displacement of mix that cannot be remedied in the final rolling.

Pneumatic-tire rolling should be continuous after breakdown rolling until all the mix placed has been thoroughly compacted. Turning of pneumatic-tire rollers on the fresh mix should only be permitted where it does not cause scuffing or undue displacement.

Some mixes exhibit what has been termed a tender zone at an intermediate temperature range of 230°–190°F (110°–88°C). This is most common with coarse-grade mixtures that may retain a slight amount of moisture after production. Rolling on the mix in this temperature zone with a steel drum will cause the mix to laterally displace or be shoved to the point of tearing the mat. If this phenomenon occurs, compaction with steel-wheel rollers should cease until the mixture is sufficiently stable to support the roller without shoving. Pneumatic-tire rollers, which apply less contact pressure than steel-wheel rollers, are less likely to seriously deform the surface under these conditions. Additional densification can also be achieved with the finish roller if it is sufficiently heavy and operated slowly.

Vibratory rollers—of proper static weight, vibration frequency, and amplitude—are used to meet required densities with fewer coverages than with static-weight tandem or pneumatic-tire rollers (or combinations of the two). Regardless of the type of rollers used, the rolling pattern should be developed in the same manner as for the initial, or breakdown, rolling. This pattern should be continued until the desired density is obtained.

Finish Rolling

Finish rolling is done primarily for the aesthetic improvement of the surface. It should be accomplished with steel-wheel tandems operating in static mode (without vibration) while the material is still warm enough for removal of roller marks.

Compaction is the last step that will determine the performance of the HMA. There is a seemingly endless variety of roller patterns to accomplish this task. It is a critical function, and finding the right pattern for producing consistent density results while maintaining smoothness and texture is a rewarding and long-lasting effort.

9.4.6 Rolling Joints and Edges

Transverse Joints

Just as a variety of different roller patterns are available, a number of different techniques are also available for compacting a transverse joint. Unless there are restrictions on either side of the lane being placed (live traffic, barriers, curbs, etc.), most operators like to cross roll the joint with the roller on the previously laid material, except for a 6-inch (150 millimeters) projection of the wheels when using a tandem (see Figure 9.12) and of one rear wheel when using a three-wheel roller. This operation should be repeated with successive passes covering 6–8 inches (150–200 millimeters) of fresh material until the entire width of a drive wheel is on the new mix.

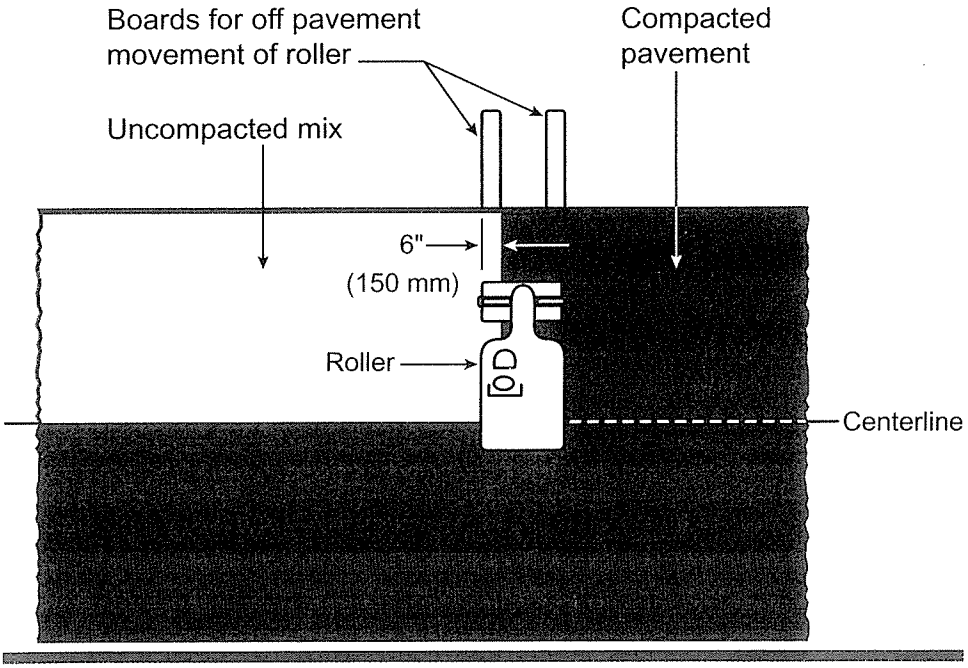


Figure 9.12 Rolling a Transverse Joint

Boards of proper thickness should be placed at the edge of the pavement to provide for off-the-pavement movement of the roller. The thickness of the board is equal to how much the HMA will compact or roll down—normally 20–25 percent. These "rolldown" factors will vary depending on mix type. SMA/gap-graded mixtures and open-graded mixtures will roll down less than dense-graded mixtures. If boards are not used, transverse rolling must stop 6–8 inches (150–200 millimeters) short of the outside edge in order to prevent damage to the edge. The outside edge then must be rolled out later during longitudinal rolling.

If lane restrictions prohibit cross rolling, it is possible to roll straight onto the new mat very slowly so that the mix will be pinched down into the joint area instead of being pushed away from it. Other options include pinching approximately 6 inches (150 millimeters) by transverse rolling, and then rolling in straight. The goal is to make a tight, level joint flush with the existing edge. This joint should be checked with a straightedge, typically before, during, and after rolling to ensure that it meets specifications.

Longitudinal Joints

Just as there are options for rolling transverse joints, options for rolling longitudinal joints also exist. The method chosen should be maintained to ensure uniformity. Longitudinal joints should be rolled directly behind the paving operation as illustrated in Figure 9.13.

Hot-side rolling. For lifts under 3 inches, it is best to roll from the hot side and place most of the roller on the hot mat with only about 6 inches (150 millimeters) of the drum overlapping on the adjacent cold mat. This approach places the roller on the uncompacted mat, and the roller is both compacting the mix and the joint at the same time. This is also the only realistic way to roll a longitudinal joint when paving with traffic present in the adjoining lane. This is generally more efficient than cold-side rolling.

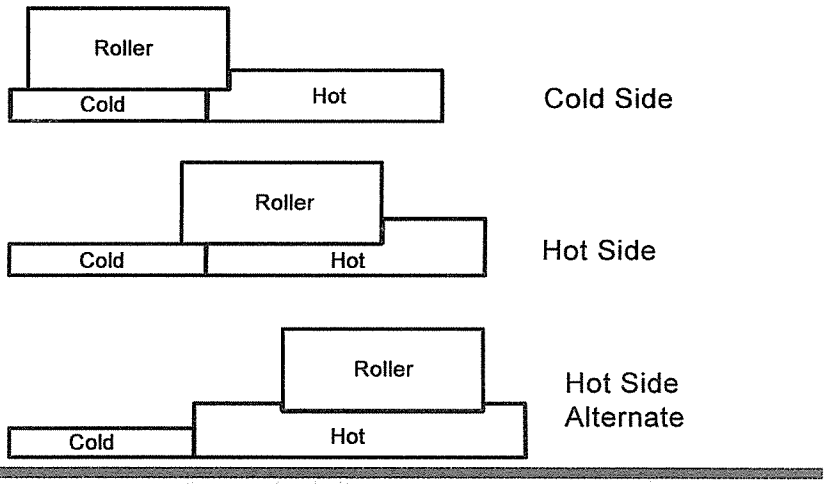


Figure 9.13 Options for Longitudinal Joint Compaction